

Scaling Databases

Software System Development
Monsoon 2026

Structured Query Language

Define and Manage data

CREATE, ALTER, DROP, TRUNCATE, RENAME, COMMENT

Manipulates data

SELECT, INSERT, UPDATE, DELETE, MERGE, CALL, EXPLAIN PLAN, LOCK TABLE

Organize data

CTE, SUB QUERIES, STORED PROCEDURES, CURSORS, USER DEFINED FUNCTIONS, TRIGGERS, INDEXES, VIEWS, MATERIALIZED VIEWS, ACID CONTROLS, WINDOW FUNCTIONS

Execution Plan

When you submit a SQL query, the database engine processes the query through following stages

Parse → Optimize → Execute

- Operators - Table Scan, Clustered Index Scan, Index Seek, Nested Loops, Merge Joins, and Hash Joins.
- Data Flow Direction
- Cost Metrics
- Cardinality (Row Estimates vs. Actuals)

EXPLAIN SELECT * FROM Employees;

Estimated Execution Plan – *Runs before query execution*

Actual Execution Plan – *Report generated after completion of query execution*

Data Extraction Approaches

- BULK INSERT
- LOAD DATA INFILE
- COPY

```
BULK INSERT Employees
FROM 'C:\Data\employees.csv'
WITH (
    DATAFILETYPE = 'char',
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    FIRSTROW = 2,           -- Skips the header row in the CSV
    BATCHSIZE = 5000,       -- Imports in chunks of 5,000 rows
    KEEPNULLS,              -- Preserves empty fields as NULL
    TABLOCK                 -- Optimizes performance via table-level locking
);
```

```
LOAD DATA INFILE '/var/lib/mysql-files/employees.csv'
INTO TABLE Employees
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

```
COPY Employees FROM '/tmp/employees.csv' WITH (FORMAT csv, HEADER true);
```

Data Transformation Strategies

- **String Transformation**

- **Case Conversion:** LOWER(col), UPPER(col), and INITCAP() equivalents.
- **Trimming Whitespace:** TRIM(col), LTRIM(col), and RTRIM(col) to remove unwanted spaces from raw files.
- **Substring & Extraction:** SUBSTRING(col, start, length), LEFT(), and RIGHT() to slice text.
- **Replacing Values:** REPLACE(col, 'search_string', 'replace_string') to clean up typos or standardize formatting (e.g., changing shorthand states to full names).

- **Numerical Transformation**

- **Rounding & Truncation:** ROUND(col, decimals), FLOOR(col), and CEIL(col).
- **Absolute Values & Modulo:** ABS(col) and MOD(col, divisor).
- **Condition-Based Math:** Performing conditional arithmetic directly using CASE expressions.

- **Date Transformation**

- **Parsing & Formatting:** DATE_FORMAT(col, '%Y-%m-%d') and STR_TO_DATE(string_col, '%d/%m/%Y') to convert unstructured date strings into true SQL DATE or DATETIME types.
- **Date Arithmetic:** DATE_ADD(col, INTERVAL 1 MONTH), DATEDIFF(date1, date2), and TIMESTAMPDIFF() to calculate ages, tenure, or date spans.

Data Load Strategies

- **Full Load (Truncate and Load)** - Every time data is ingested, the existing target table is completely emptied (truncated) and refilled with the entire dataset from scratch.
- **Incremental Load (Append-Only)** - Only new records that have been created since the last load are appended to the target table. Existing historical data remains untouched.
- **Upsert / Merge Strategy (Insert or Update)** - The system checks incoming records against existing records in the database. If the record already exists (usually matched via a primary key or unique identifier), it is updated; if it does not exist, it is inserted.
- **Staging and Batch Load Strategy** - Raw data is first loaded into a temporary staging table (which has minimal constraints or indexes). Once inside the database, transformations and validations are applied via SQL, and the data is inserted into production tables in optimized batches to avoid locking up production resources.
- **Partitioned Table Swapping (Rolling Load)** - Data is loaded into a separate, newly constructed staging table or a separate table partition. Once the data load and indexing are complete, the partition is instantly swapped into the main table structure with minimal downtime.

Designing Datawarehouse

- Star Schema
- Snowflake Schema

***fact table** stores quantitative, numerical measurements (or "metrics") resulting from a business process or transaction.*

***dimension table** stores descriptive attributes that provide context to the quantitative measurements stored in fact tables.*

